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# Intellectual achievement and cognitive functioning in children and adolescents with ADHD-affected siblings: A systematic literature review and meta-analysis

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ABSTRACT

*Background:* Emerging literature suggests that neurocognitive traits commonly observed in children and adolescents with attention-deficit/hyperactivity disorder (ADHD) are present, albeit to a lesser degree, in their non-ADHD siblings. This meta-analytic review aimed to quantitatively summarize intellectual and cognitive performance differences between children and adolescents without ADHD and both their ADHD-diagnosed siblings and unrelated peers without ADHD.

*Methods:* We implemented strategic search algorithms in four indexing databases (PubMed/MEDLINE, Web of Science, PsycINFO, and ProQuest) to retrieve potentially eligible reports published up to 5th June 2025. Meta-analyses were conducted using three-level hierarchical models to account for the dependency among effect sizes derived from the same study. Between-group differences in intellectual and cognitive scores were quantified using standardized mean differences (SMD). Moreover, we assessed the robustness of the findings by testing for potential small-study effects.

*Results:* Based on 404 unique effect sizes derived from 32 studies, involving a total of 8873 participants, the current results demonstrate that the overall cognitive-intellectual performance of siblings surpasses their ADHD-affected siblings (SMD = 0.30, 95 % CI [0.23, 0.38]), but falls short of unrelated peers (SMD = −0.28, 95 % CI [−0.35, −0.21]). This suggests that non-ADHD siblings exhibit an intermediate neurocognitive profile between their ADHD-affected siblings and peers without ADHD. Notably, we observed this pattern of results consistently across the major cognitive categories examined (intellectual functioning, attention and perceptual-temporal processing, memory, and executive functions). Importantly, no evidence of small-study effects was found.

*Conclusion:* The current findings provide robust support for cognitive-intellectual vulnerabilities in siblings of ADHD-diagnosed children and adolescents, possibly accounted for by genetic inheritance, shared environmental influences, and gene-environment interactions.

## 1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a childhood-onset neuropsychiatric condition characterized by non-adaptative (developmentally age-inappropriate) levels of inattention, impulsivity, and/or hyperactivity, often accompanied by neuropsychological (e.g., difficulty sustaining attention when managing and completing school-work) and social (e.g., tendency to interact in a manner that does not lead to peer acceptance) difficulties (American Psychiatric Association, 2013). Based on over three decades of family studies, the heritability of ADHD has been estimated at 74 % (Faraone & Larsson, 2019), and a previous systematic review documented that the prevalence of ADHD among siblings of children and adolescents affected by the condition ranges from 26 % to just over 45 % (Hidalgo-López et al., 2019). This is in line with past research suggesting that siblings of children and adolescents with ADHD are at a ninefold increased risk of developing ADHD

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compared to control siblings (Chen et al., 2008). From a genetic standpoint, biological siblings of probands with ADHD share on average 50 % of their genetic information, and hence non-proband siblings are vulnerable to ADHD-related neurobiological characteristics, including the developmental structure of genetic risk factors that contribute to ADHD-related cognitive difficulties. This is particularly relevant in light of accumulating evidence indicating that ADHD-diagnosed children and adolescents exhibit inferior test performance relevant to a number of important cognitive processes such as working memory (Ramos et al., 2020), time perception (Marx et al., 2022), and response inhibition (Lipszyc & Schachar, 2010), in addition to higher chances of academic underachievement (Loe & Feldman, 2007). Collectively, these findings provide strong motivation to profile cognitive and intellectual performance in siblings without ADHD to determine any difficulties.

The literature on cognition in siblings of ADHD-affected children and adolescents has gained increasing attention in recent decades but has yielded inconclusive findings. While some studies suggest that non-ADHD siblings perform similarly to peers without ADHD on most neuropsychological measures (Durston et al., 2004; Fan et al., 2018; Goos et al., 2009), a growing body of evidence indicates that siblings of ADHD-diagnosed children and adolescents share neurocognitive vulnerabilities (Kuntsi et al., 2010), including significant difficulties in neuropsychological measures designed to assess cognitive inhibition, working memory, and intellectual functioning (Albrecht et al., 2010; Bidwell et al., 2007; Doyle et al., 2005; Nikolas & Nigg, 2015; Rommelse, Altink, Oosterlaan, Buschgens, et al., 2008). These mixed findings may be attributed to differences in study designs, including variations in the cognitive-intellectual measures employed across studies. Another factor that may be driving these inconsistencies relates to how siblings are deemed to not have ADHD. In scientific literature, substantial variability exists across studies in how this classification is operationalized, and in many cases, the extent of behavioral characteristics below the diagnostic threshold (i.e., subthreshold ADHD traits) is not clearly reported, thereby preventing analysis. Throughout this meta-analytic synthesis, therefore, “unaffected siblings” and “non-ADHD siblings” refer to child or adolescent biological siblings who did not meet full diagnostic criteria for ADHD.

Importantly, an evidence-based understanding of the intellectual and cognitive profile of siblings of ADHD-diagnosed children and adolescents holds critical implications for both research and clinical practice. By aggregating data across studies, a meta-analytic approach helps clarify whether non-ADHD siblings exhibit an intermediate neurocognitive profile between ADHD probands and unrelated peers without ADHD, as suggested by some previous research (e.g., Faraone et al., 1993; Gau & Shang, 2010; Leon et al., 2024; Rommelse, Altink, Oosterlaan, Buschgens, et al., 2008). Furthermore, as individual studies often report inconsistent findings due to variations in design, sampling, statistical power and measurement tools (Button et al., 2013; Harris & Smith, 2009; Ramos et al., 2020), the meta-analytic approach offers a robust strategy to (at least partially) address such inconsistencies by examining whether pre-specified moderators (e.g., age, sex, cognitive domain) contribute to the heterogeneity of the pooled effect sizes. The outcomes have stronger potential to inform more clearly on intellectual-cognitive performance differences in siblings without ADHD, thus providing invaluable information relevant to early identification of at-risk family members (e.g., by informing early cognitive screening protocols) and for guiding support services and educational programs for caregivers and stakeholders.

Considering all mentioned above, this quantitative review aimed to meta-analyze intellectual and cognitive performance differences in the following groups of children and adolescents: (a) unaffected vs. ADHD-diagnosed siblings; and (b) non-ADHD siblings vs. unrelated controls without ADHD. In addition, we explored potential moderators that may play critical roles in between-group performance differences by means of subgroup and meta-regression analyses. Note that we initially also intended to investigate the neurocognitive profile of unaffected offspring of ADHD-affected individuals, but no study meeting the eligibility criteria was found.

## 2. Methods

### 2.1. Guideline and data availability

The present research synthesis was reported in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). In the spirit of Open Science, statistical coding and spreadsheets containing the collected data are available at the Open Science Framework (https://osf.io/sjwf3/). Although this systematic review was not pre-registered, and hence no protocol is available, all procedures (e.g., search strategies, eligibility criteria, and analytic methods) were conducted according to established meta-analytic standards, were fully reported to support reproducibility, and investigated a priori research questions that were theoretically grounded in the literature.

### 2.2. Eligibility criteria

Randomized controlled trials and cross-sectional, cohort, and case-control studies published in English, Spanish, or Portuguese were eligible for inclusion. In addition, studies had to meet the following criteria. First, the diagnosis of participants with ADHD had to be based on established diagnostic criteria according to the Diagnostic and Statistical Manual of Mental Disorders (DSM). Second, consistent with a previous ADHD meta-analysis (Vertessen et al., 2022), we considered only studies reporting on at least one intellectual or cognitive outcome for children and adolescents aged $\geq 5$ and $\leq 18$. Third, studies had to report the statistics needed to calculate standardized effect sizes (e.g., sample size, mean and standard deviation). In cases where a study provided only variance-based effect sizes (e.g., partial eta-squared, Cohen’s *f*) for between-group differences, we converted these statistics to standardized mean differences (SMD) before quantitatively summarizing results. Fourth, all meta-analyzable outcomes had to be derived from objective neurocognitive testing, either using validated measurement tools for cognition and/or academic achievement (e.g. Digit Span, Peabody Individual Achievement Test [PIAT]), or in cases of non-standard measures, the study had to detail the test design and the methodology followed in administering the test (or cite the relevant reference containing these details); if not, the measure was excluded from the meta-analyses. Studies relying solely on questionnaires or rating scale data (e.g., parent or teacher reports) were not considered in the quantitative syntheses. Moreover, we excluded publications based on the following additional criteria: duplicated literature, single case studies, conference proceedings and abstracts, and qualitative or theoretical research.

### 2.3. Information sources and search strategy

The systematic literature search was initially carried out on 13$^{th}$ December 2022, in four indexing databases (PubMed/MEDLINE, Web of Science Core Collection, PsycINFO, and ProQuest), without applying date restrictions. Subsequently, we updated this literature search on 30$^{th}$ June 2024 and further on 5$^{th}$ June 2025 using the same search terms. In addition, the reference lists of articles deemed relevant were cross-checked for further eligible studies. In the Supplementary Online Content, the Appendix details the search algorithms (strategic syntax) for each database. The search string for cognitive outcomes implemented in all databases stemmed from a comprehensive umbrella review (Pievsky & McGrath, 2018) that provided an extensive list of cognitive measures commonly administered in ADHD research.

### 2.4. Selection process

EndNote 20 (Thomson Reuters) software managed citations and removed duplicate reports. Unique references were uploaded to the online platform Ryyan (www.rayyan.ai) for title/abstract and full-text screening. The third and fourth authors independently screened the retrieved reports for eligibility; in cases of disagreement at the level of title/abstract screening, a consensus was reached after discussion with the first author. The first and second authors assessed the full-texts and determined the final study eligibility.

### 2.5. Data collection process

Based on a pre-defined Microsoft Excel template, we extracted the following information from each included study: the country where the study took place, first author, year of publication, study design (e.g., cross-sectional or cohort), demographic data (percentage of boys, and sample mean age and range), and the meta-analyzable statistics from each included outcome measure (e.g., sample size, mean and standard deviation, *t* or *F* test results). For children and adolescents with ADHD, we further collected information on methylphenidate pharmacology (coded as naïve, else medicated or not medicated during the cognitive testing), ADHD presentation (combined, predominantly inattentive, or predominantly hyperactive/impulsive), and diagnosis criteria (i.e., edition of DSM). For longitudinal studies with follow-up assessments, only baseline test scores were extracted. Importantly, the first author collected data from the included reports, and the second author cross-screened for inconsistencies and double-checked all extracted data to ensure accuracy.

### 2.6. Population, intervention/exposure, comparator, and outcome

The population of interest consisted of ADHD-unaffected children and adolescents with a child or adolescent sibling diagnosed with ADHD. For the purposes of this quantitative synthesis, any intervention or exposure involving cognitive testing that met the inclusion criteria was considered eligible. As previously noted in the Introduction, we were interested in two comparisons: (a) unaffected vs. ADHD-affected siblings (Meta-Analysis I); and (b) ADHD-unaffected siblings vs. unrelated peers without ADHD (Meta-Analysis II). The measured outcomes of interest were the average scores on intellectual and/or cognitive tests for each of the compared groups.

### 2.7. Sample overlap

In the present quantitative review, we carefully analyzed test scores that came from the same study population. In cases of two or more publications with overlapping sample, we included the test scores from the study with the largest sample size (given the same intellectual/cognitive test) and test scores from unique intellectual/cognitive tests. This approach allowed us to meta-analyze a single effect size for each test from the included studies. To achieve this, we initially grouped studies with sampling overlap into three subgroups. The first subgroup comprised 27 potentially eligible studies that utilized data from the International Multi-Center ADHD Genetics Project (IMAGE), a research initiative conducted in seven European countries (Belgium, Germany, Netherlands, Ireland, Spain, Switzerland, and United Kingdom) and Israel between 2003 and 2006. In addition, we identified two studies from the NeuroIMAGE project (2009–2012), a follow-up cohort study that collected cognitive test scores from participants at the Dutch site of the IMAGE project. From this subgroup, 10 studies (Albrecht et al., 2008; Banaschewski et al., 2012; Bitsakou et al., 2009; Christiansen & Oades, 2010; Himpel et al., 2009; Jiang et al., 2020; Oades & Christiansen, 2008; van der Stigchel et al., 2007; van Lieshout et al., 2019; Wood et al., 2011) were retained in the quantitative analyses. The second subgroup consisted of nine overlapping sample studies conducted at the National Taiwan University Hospital. Following our analysis, five studies (Fan et al., 2018; Faraone et al., 1993; Gau & Huang, 2014; Gau & Shang, 2010; Hwang-Gu & Gau, 2015) from this subgroup were included in the current meta-analyses. The third subgroup encompassed three studies (Slaats-Willemse et al., 2003; Slaats-Willemse et al., 2005; Slaats-Willemse et al., 2007) conducted by Slaats-Willemse et al., and all were retained since they employed different neuropsychological measures (albeit involving the same participants).

### 2.8. Cognitive domain

The full list of tests and subtests considered is provided in Table 1. To organize the wide range of intellectual and neuropsychological outcomes included in this quantitative review, we employed different approaches to classify each test outcome according to a primary neuropsychological domain (e.g., verbal memory, cognitive flexibility) – here referred to as *cognitive subdomains*. For Performance IQ and Verbal IQ Indexes, both derived from previous editions of the Wechsler Intelligence Scale for Children (WISC), we followed the classification summarized in the Encyclopedia of Clinical Neuropsychology (2011). For well-known and widely used tests (e.g., Wisconsin Card-Sorting Test), we followed Strauss et al. (2006). For the cognitive measures that comprise the Cambridge Automated Neuropsychological Test Battery (CANTAB), we employed the classification available on the CANTAB website (https://cambridgecognition.com/digital-cognitive-assessments/). In cases of non-standard measures from studies that met the inclusion criteria, we relied on detailed test descriptions or referenced publications to determine an appropriate classification. This framework yielded 12 cognitive subdomains (see Table 1).

The subdomains were used to structure the subgroup analyses and the presentation of results. Although low statistical power in subgroup analyses is a common drawback in meta-analyses, there is no rule of thumb to determine how many studies should be considered in such analyses (Cuijpers et al., 2021; Jackson & Turner, 2017; Myles et al., 2021). To minimize the risk of power-related bias in interpreting results from subgroup analyses, in the present meta-analytic review we grouped the 12 subdomains into four major *cognitive domains* as follows:

1) The *Intellectual Functioning* domain includes tests and subtests designed to measure global intelligence (e.g., Full-Scale IQ and estimated IQ from the Wechsler Intelligence Scale for Children [WISC]), scholastic achievement (e.g., Wide Range Achievement Test), Performance IQ (e.g., Block Design and Object Assembly from the WISC), and Verbal IQ (e.g., Arithmetic and Similarities from the WISC).
2) The *Attention and Perceptual-Temporal Processing* domain comprises (sub)tests relevant to the assessment of attention and psychomotor speed (e.g., Part A of the Trail Making Test, Continuous Performance Test) and perceptual-temporal processing (e.g., Motor Timing Task, Time Reproduction Task).
3) The *Memory* domain was derived from only four studies (Gau & Shang, 2010; Luca et al., 2007; Nikolas & Nigg, 2015; Shang & Gau, 2011) that provided scores on measures designed to assess verbal memory (e.g., Digit Span Forwards) and visuospatial memory (e.g., Delayed Matching to Sample from CANTAB), and hence the summary effect size from this major cognitive domain should be interpreted with caution. The limited number of studies and the availability of data precluded a more detailed classification into specific subdomains (e.g., short-term memory, episodic memory).
4) The *Executive Function* domain is an umbrella category that encompasses a range of tasks purported to assess the constructs of cognitive flexibility (e.g., Wisconsin Card Sorting Test), cognitive inhibition (e.g., Go/No-Go Task, Stop-Signal Task), working memory (e.g., Digit Span Backwards), and planning (indexed only by the Stockings of Cambridge from CANTAB).

**Table 1**
Tests and subtests included in the meta-analyses according to the cognitive domain and subdomain.

| Cognitive *subdomains* grouped by four major cognitive *domains* | Outcome Measure |
|---|---|
| **Intellectual Functioning** | |
| Global Intelligence | Estimated IQ (WISC-R, WISC-III, WISC-IV) |
| | Estimated IQ – Grundintelligenztest Skala 2 (CFT 20-R) |
| | Full-Scale IQ (WISC, WISC-R, WISC-III) |
| Scholastic Achievement | Peabody Individual Achievement Test (PIAT) ( Dunn & Markwardt, 1970) |
| | Wide Range Achievement Test (WRAT) |
| | Woodcock Reading Mastery Test (WRMT) |
| Performance IQ | Block Design (WISC, WISC-R, WISC-III) |
| | Coding (WISC, WISC-R) |
| | Object Assembly (WISC) |
| | Performance IQ Index (WISC, WISC-R, WISC-III) |
| | Picture Arrangement (WISC) |
| | Picture Completion (WISC) |
| Verbal IQ | Arithmetic (WISC, WISC-R) |
| | Comprehension (WISC) |
| | Information (WISC) |
| | Similarities (WISC) |
| | Verbal IQ Index (WISC, WISC-R, WISC-III) |
| | Vocabulary (WISC, WISC-R, WISC-III) |
| **Attention and Perceptual-Temporal Processing** | |
| Attention and Psychomotor Speed | Amsterdam Neuropsychological Tasks (De Sonneville, 1999) – Pursuit Task |
| | Amsterdam Neuropsychological Tasks (De Sonneville, 1999) – Sustained Attention Task |
| | Amsterdam Neuropsychological Tasks (De Sonneville, 1999) – Tracking Task |
| | CANTAB - Match to Sample Visual Search (MTS) |
| | CANTAB - Reaction Time (RTI) |
| | CANTAB - Rapid Visual Information Processing (RVP) |
| | Composite Score of Processing Speed[§] |
| | Continuous Performance Test (CPT) (Gau & Huang, 2014) |
| | Quantified Behavior Test (QbTest) (Ulberstad, 2012) |
| | Symbol Search (WISC-III) |
| | Trail Making Test – Part A |
| Perceptual-Temporal Processing | Composite Score of Temporal Processing[§] |
| | Duration Discrimination Tasks (Rammsayer & Lima, 1991; Rammsayer & Ulrich, 2001) |
| | Motor Timing Task (Rommelse, Altink, Oosterlaan, Beem, et al., 2008; Van Meel et al., 2005) |
| | Perceptual Speed Composite Score[f] |
| | Time Estimation Task (Barkley et al., 1997) |
| | Time Reproduction Task (Hwang-Gu & Gau, 2015) |
| | Verbal Estimation Task (Hwang-Gu & Gau, 2015) |
| **Memory** | |
| Verbal Memory | Composite Score of Memory[§] |
| | Comprehensive Test of Phonological Processing (CTOPP) (Wagner et al., 1999) |
| | Digit Span Forwards (WISC, WISC-III) |
| Visuospatial Memory | CANTAB – Delayed Matching to Sample (DMS) |
| | CANTAB – Paired Associates Learning (PAL) |
| | CANTAB – Pattern Recognition Memory (PRM) |
| | CANTAB – Spatial Recognition Memory (SRM) |
| **Executive Function** | |
| Cognitive Flexibility | CANTAB – Intra-Extra Dimensional Set Shift (IED) |
| | Rapid Automatized Naming (Denckla & Rudel, 1976) |
| | Shifting Attentional Set Task (Slaats-Willemse et al., 2007) |
| | Switch-Task (Cepeda et al., 2000; Oades & Christiansen, 2008) |
| | Trail Making Test – (Part B or Part B-A) |
| | Wisconsin Card-Sorting Test (WCST) |
| Cognitive Inhibition | Composite Score of Cognitive Inhibition[§] |
| | Flanker Task (Kopp et al., 1996) |
| | Go/No Go Task |
| | Oculomotor Capture Task (van der Stigchel et al., 2007) |
| | Stop-Signal Task |
| | Stroop Test |
| Planning | CANTAB – Stockings of Cambridge (SoC) |
| Working Memory | CANTAB – Spatial Working Memory (SWM) |
| | Composite Score of Working Memory[§] |
| | Counting Span (Case et al., 1982) |
| | Digit Span – Forwards plus Backwards (WISC, WISC-R) |
| | Digit Span Backwards (WISC-III) |
| | Freedom from Distractibility Index (WISC-R) |
| | Sentence Span (Siegel & Ryan, 1989) |
| | The Fast Task (Andreou et al., 2007; Kuntsi et al., 2005; Wood et al., 2011) |
| | Visuospatial Working Memory Task (Nutley et al., 2010) |

Abbreviations: IQ, intelligence quotient; WISC, Wechsler Intelligence Scale for Children (R, revised; III, third edition; IV, fourth edition); CANTAB, Cambridge Automated Neuropsychological Test Battery.

[§] Nikolas and Nigg (2015) developed the following composite scores for their sample: (a) Composite Score of Cognitive Inhibition: Delis-Kaplan Executive Function System (DKEFS) Colour-Word Inhibition Time, DKEFS Inhibition/Switching Time, and Stop Task Stop Signal Reaction Time; (b) Composite Score of Memory: Spatial Span Forward and Digit Span Forward; (c) Composite Score of Processing Speed: DKEFS Colour-Word Reading Time and DKEFS Trail Making Number Sequencing Time; (d) Composite Score of Temporal Processing: Tapping Task Visual 400-ms detrended SD, Tapping Task Auditory 400-ms detrended SD, Tapping Task Visual 1000-ms detrended SD, and Tapping Task Auditory 1000-ms detrended SD; and (e) Composite Score of Working Memory: Spatial Span Backward, Digit Span Backward and DKEFS Trail Making Number Sequencing Time.

[f] Based on scores from the Educational Testing Service Identical Pictures Test and Colorado Perceptual Speed.

Note that the systematic review identified two studies (Bidwell et al., 2007; Wood et al., 2011) that assessed delay aversion, a construct used in the ADHD literature to characterize motivational style (Fernández-Martín et al., 2025; Sonuga-Barke, 2002, 2003). Given the cognitive/intellectual focus of this review, outcomes related to delay aversion were not included in the present quantitative synthesis.

### 2.9. Effect size direction

Note that we carefully aligned the direction of the average scores before conducting the statistical analyses because the direction (interpretation) of treatment effects differs across various neuropsychological measures. For instance, scores on the Trail Making Test reflect the time spent to complete the test, with higher scores indicating poorer performance. In contrast, scores on Digit Span Backward reflect the number of items correctly recalled, with higher scores indicating better performance. To aid interpretability of the current results, we computed effect sizes such that positive values reflect better cognitive performance. In the Supplementary Online Content, eTable 1 summarizes the effect size, along with the variance and standard error, for each intellectual/achievement and neuropsychological measure considered for inclusion in the two meta-analyses.

### 2.10. Risk of quality-related bias in individual studies

We employed the checklists for analytical cross-sectional, case-control, and cohort studies from the Joanna Briggs Institute (JBI) Critical Appraisal Tools (Moola et al., 2020) to rate the risk of quality-related bias in the individual studies included. Note that no randomized controlled trial met the inclusion criteria. Following best practice criteria related to research design, methodology, and statistical analyses, the JBI templates allow studies to be categorized into levels of quality based on the allotted response for each question (yes, no, unclear, or not applicable [NA]). Consistent with a recent meta-analysis with a design analogous to the current study (Ramos et al., 2022), we used the following cut-off values to rate study quality: $\leq$ 49 % yes responses, high risk of bias; 50–69 % yes responses, moderate risk of bias; $\geq$ 70 % yes responses, low risk of bias. Note that the second author graded the quality of each study, and the first author cross-checked for inconsistencies. Since all but one of the studies met the criteria for low risk of bias (indicating high quality), we did not conduct subgroup analyses including study quality as a moderator variable.

### 2.11. Age differences

In the Supplementary Online Content, eTable 2 shows a detailed study characterization based on the age of the included samples in each group. In both the ADHD and control groups, the sample mean age ranged from 6 to 18 years, whereas in the unaffected sibling group the sample mean age ranged from 5 to 18 years. Note that we initially planned to classify participants in each group as either children ($5 \leq$ sample mean age $\leq 12$) or adolescents ($13 \leq$ sample mean age $\leq 18$) to determine potential effect size differences between these two age categories in subgroup analyses. However, only two studies (Bitsakou et al., 2009; Halperin & Gittelman, 1982) provided sufficient data to categorize the included participants as either children or adolescents. Hence, we did not perform subgroup analyses by age category and instead included the sample mean age as a continuous predictor covariate in meta-regression analyses.

### 2.12. Statistical analyses and synthesis methods

We performed all the statistical analyses in R (R Core Team, 2022) with R Studio (RStudio Team, 2022). To control for the within-study dependency of effect sizes (correlated outcomes), the *metafor* package fitted an overall three-level meta-analysis and three-level subgroup and meta-regression analyses for each comparison group, using restricted maximum likelihood estimation and random-effects modeling with the *rma.mv* function. To improve the robustness of our inference and account for potential model misspecification, we applied cluster-robust variance estimation (CRVE) in all fitted meta-analytic models (Hedges et al., 2010). Based on the formula devised by Hedges (Hedges, 1981), the *escalc* function of the *metafor* package computed standardized mean differences (SMDs) correspondent to the Hedges' *g* effect size – an unbiased variance estimator for meta-analyses including small-sized samples ($N \leq 20$). Moreover, we plotted the standardized residuals from the three-level meta-analyses to identify influential data points (outliers) that can distort the effect size precision. Residuals $< -1.96$ or $> 1.96$ were deemed influential outliers, given that we would expect that 95 % of the residual values would fall within this range under a standard normal distribution (Kutner et al., 2004). All R scripts are available online at https://osf.io/sjwf3/.

To characterize the extent of variation in effect sizes, we calculated two indexes of heterogeneity not accounted for by sampling error: $\tau^2$, an estimation of absolute effect size dispersion, and $I^2$, an estimation of the average percentage of variation in effect sizes. Given the implementation of multilevel analyses, we reported on two levels of random-error-free heterogeneity: $\tau_2^2$ and $I_2^2$ for the within-study/cluster variance (level 2) and $\tau_3^2$ and $I_3^2$ for the between-study/cluster variance (level 3). In the current quantitative review, $I^2$ statistics of 25 %, 50 %, and 75 % were interpreted as suggesting low, moderate, and substantial levels of heterogeneity, respectively (Higgins & Thompson, 2002). Subgroup analyses investigated the influence of the four major cognitive domain category clusters (intellectual functioning, attention and perceptual-temporal processing, memory, and executive functions) on the effect size variance, and the Wald-type test assessed potential statistically significant differences in effect sizes across the major cognitive domain categories. Similarly, univariate meta-regression analyses examined whether numerical demographic data (sample mean age and percentage of boys) and year of publication moderate between-group performance differences. In meta-regression analyses, we applied listwise deletion to handle missing data.

### 2.13. Small-study effects

In the meta-analytic literature, small-study effects are a type of publication bias and reflect the tendency for small-sized studies to estimate different (typically larger) treatment effects than studies with larger samples (Schwarzer et al., 2015). To address this issue, in the current quantitative review we first visually inspected the graphical presentation of the data in funnel plots. In the absence of small-study effects, data points (the SMD of each study) are expected to be roughly symmetrically distributed in upside-down funnel plots. To statistically ascertain the risk of small-study effects, we applied a well-known graph-based method of funnel plot asymmetry (Egger's regression test) and a selection model method (the three-parameter selection model [3PSM]) (Coburn & Vevea, 2019; Vevea & Hedges, 1995). The 3PSM assumes that studies are not randomly selected and allows estimation of the impact of potential missing studies. Note that we implemented the weight-function model in R using its default two-tailed significance threshold of $p = .05$ to distinguish between significant and non-significant effects.

It is important to mention that both Egger's regression test and the 3PSM assume independent effect sizes and do not account for the complex hierarchical structure of the data. To partially mitigate this limitation, we applied these tests to study-level pooled effect sizes. The number of effect sizes per study ranged from 1 to 24 (median = 6) in both Meta-Analysis I and II, as most studies reported multiple intellectual or cognitive outcomes. To calculate study-specific pooled estimates, we first performed a three-level subgroup analysis by study ID (i.e., within clusters of correlated effects at the study level) and then we applied CRVE, as detailed in the R code available at https://osf.io/sjwf3/. This approach yielded a single summary effect size per study, preserved the hierarchical structure of the data, and informed the construction of both forest and funnel plots.

## 3. Results

### 3.1. Study selection

Fig. 1 illustrates the major steps of the systematic review. The literature search in online databases, in addition to the citation searching, identified 3668 publications, of which only 32 studies met the eligibility criteria for inclusion in the quantitative syntheses. Importantly, eTable 3 (Supplementary Online Content) lists the specific reasons for excluding 176 studies initially deemed eligible during the screening phase.

### 3.2. Study characteristics

Table 2 summarizes the relevant characteristics of each included study. Three studies included participants from two or more European countries, and the remaining 29 studies were conducted within a single country: Netherlands ($n = 8$), Germany ($n = 5$), Taiwan ($n = 5$), United States ($n = 5$), Canada ($n = 2$), Brazil ($n = 1$), England ($n = 1$), India ($n = 1$) and Israel ($n = 1$). Ten studies specified including only ADHD-affected

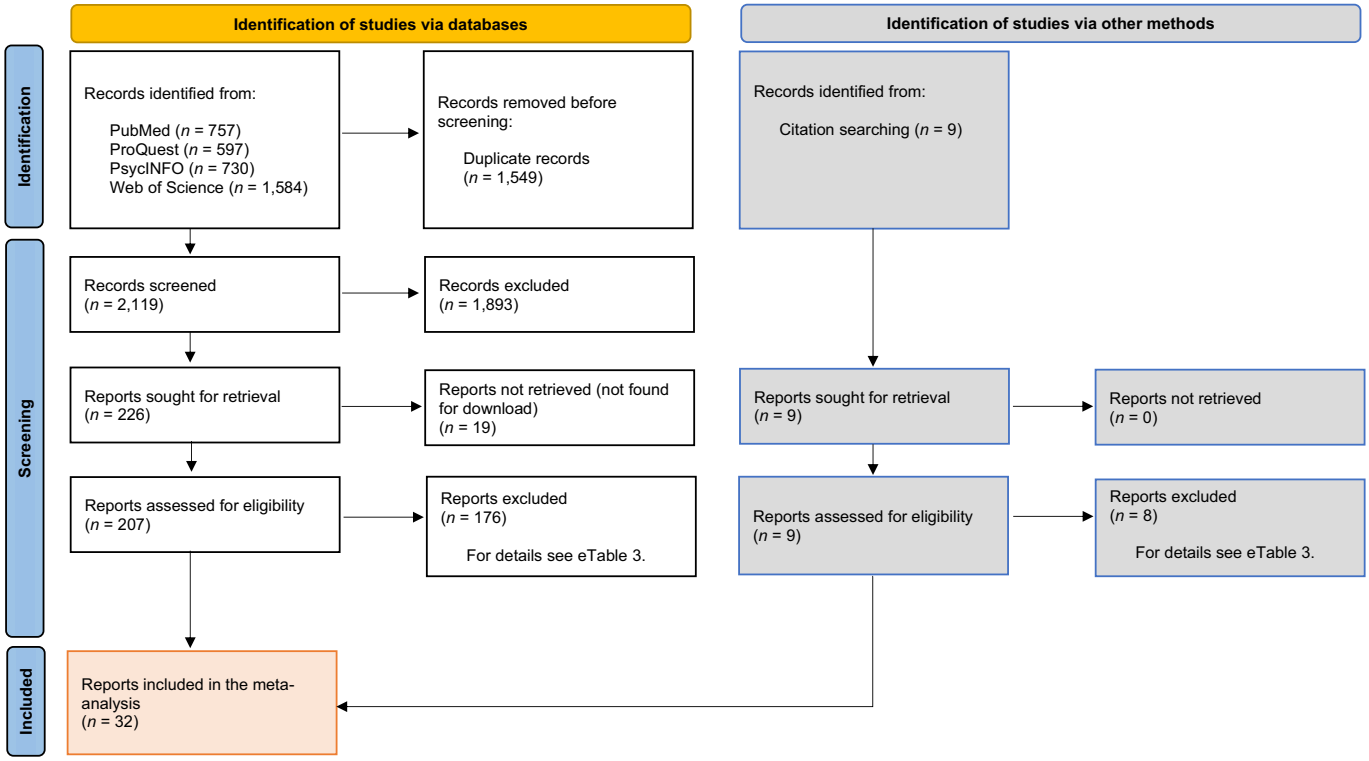



**Fig. 1.** Flow chart of study selection.

siblings with the combined presentation, and the remaining studies either reported intermixing different ADHD presentations ($n = 10$) or did not specify this information ($n = 12$). Twenty-two studies reported that the ADHD-affected siblings were in the off phase of their stimulant medication during the cognitive assessment, whereas 10 studies did not provide this information. The distribution of DSM versions used for ADHD diagnosis was as follows: DSM-5 ($n = 1$), DSM-IV ($n = 29$), DSM-III-R ($n = 1$), and DSM-II ($n = 1$). Except for two cases (Halperin & Gittelman, 1982; Oades et al., 2010), in which the applied diagnostic criteria for non-ADHD siblings could not be precisely determined, the included studies used scales based on the same DSM edition to rule out ADHD in unaffected siblings. Overall, we computed 404 unique effect sizes from the 32 studies included, with a total of 8873 participants.

### 3.3. Quality-related risk of bias in studies

As shown in Table 2, critical appraisal of the included studies revealed three types of study designs: cross-sectional ($n = 30$), prospective case-control ($n = 1$), and prospective cohort ($n = 1$). In the Supplementary Online Content, eTables 4, 5, and 6 detail the outcomes for the assessment of risk of quality-related bias in individual studies. Overall, only one study (Halperin & Gittelman, 1982) showed moderate risk of bias (moderate quality), whereas the other 31 studies met the criteria for low risk of bias (high quality).

### 3.4. Meta-Analysis I: Unaffected vs. ADHD-affected siblings

The forest plot in Fig. 2 displays the pooled SMDs for each included study and highlights the relevant outcomes for the meta-analysis that synthesized data from non-ADHD children and adolescents compared to their ADHD-affected siblings. In total, we computed 216 unique effect sizes from 31 studies, and the final meta-analysis sample consisted of 2400 non-ADHD siblings and 4007 ADHD-affected siblings. The overall pooled SMD (0.30, 95 % CI [0.23, 0.38], $p < .001$) indicates that non-ADHD siblings achieved average cognitive performance corresponding to 0.30 standard deviations above that of their siblings with ADHD. Low-to-moderate heterogeneity was observed at both level 2 ($\tau^2_2 = 0.0232$, $I^2_2 = 38.03$) and level 3 ($\tau^2_3 = 0.0219$, $I^2_3 = 35.90$). These $I^2$ values suggest that the within-study variance (level 2) was slightly higher than the between-study heterogeneity (level 3). Although we observed an outlying residual, a sensitivity analysis demonstrated that removing this outlier from the meta-analysis yielded a nearly identical SMD (0.29, 95 % CI [0.23, 0.36], $p < .001$) and had little impact on the effect size variance (level 2: $\tau^2_2 = 0.0221$, $I^2_2 = 39.38$; level 3: $\tau^2_3 = 0.0182$, $I^2_3 = 32.45$). Thus, we retained all effect sizes.

Table 3 provides the results of subgroup analyses for the outcomes of the two group comparisons we studied (unaffected vs. ADHD-affected siblings, and unaffected siblings vs. unrelated controls), specifying the cognitive subdomains falling under each of the four major cognitive domains we studied. For unaffected vs. ADHD-affected siblings, the pooled SMDs showed that children and adolescents without ADHD outperformed their ADHD-affected siblings in all four major cognitive domains we examined: intellectual functioning (0.29, 95 % CI [0.14, 0.44], $p = .001$), attention and perceptual-temporal processing (0.31, 95 % CI [0.19, 0.43], $p < .001$), memory (0.21, 95 % CI [0.04, 0.38], $p = .034$), and executive functions (0.32, 95 % CI [0.21, 0.43], $p < .001$). Group differences in all cases showed small-to-medium effect sizes. The Wald-type test ($QM_{[3]} = 1.56$, $p = .668$) indicated that there was not enough evidence to reject the null hypothesis of equal effect sizes across the four major cognitive domains analyzed. For the pooled SMDs in Table 3, within-cluster ($\tau^2_2 = 0.0234$, $I^2_2 = 37.21$) and between-cluster ($\tau^2_3 = 0.0235$, $I^2_3 = 37.46$) variance indicated low-to-moderate levels of heterogeneity.

In the Supplementary Online Content, eTable 7 displays the outcomes of the univariate meta-regression analyses. For unaffected vs. ADHD-affected siblings, the sample mean age ($\beta = -0.038$, 95 % CI [−0.105, 0.029], $p = .220$), the percentage of boys ($\beta = -0.000$, 95 % CI [−0.007, 0.006], $p = .805$) and the year of publication ($\beta = -0.010$, 95 % CI [−0.023, −0.002], $p = .080$) did not moderate the effect size heterogeneity.

**Table 2**
Characteristics of the 32 included studies and relevant clinical and demographic data.

| | | | ADHD | | | | | Siblings | | | Controls | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Study | Location | Design | Subtype | Med.[§] | Diagnosis | % Boys | Age | Non-ADHD | % Boys | Age | % Boys | Age |
| Albrecht et al. (2008) | Europe[†] | Cross-sectional cohort | Combined | Off | DSM-IV | 100 | 11.33 | DSM-IV[£] | 100 | 11.42 | 100 | 11.18 |
| Banaschewski et al. (2012) | Europe[†] | Cross-sectional cohort | Combined | Off | DSM-IV | 89 | 11.30 | DSM-IV[£] | 49 | 11.30 | 71 | 12.20 |
| Bidwell et al. (2007) | United States | Cross-sectional cohort | Mixed | Off | DSM-IV | 71 | 11.20 | DSM-IV | 37 | 11.10 | 48 | 11.90 |
| Bitsakou et al. (2009) – Sample 1 | England | Cross-sectional cohort | Combined | Off | DSM-IV | 91 | 10.72 | DSM-IV[£] | 55 | 9.45 | 59 | 10.90 |
| Bitsakou et al. (2009) – Sample 2 | England | Cross-sectional cohort | Combined | Off | DSM-IV | 86 | 14.81 | DSM-IV[£] | 48 | 14.68 | 76 | 12.15 |
| Christiansen and Oades (2010) | Germany | Cross-sectional cohort | Combined | Off | DSM-IV | 91 | 11.20 | DSM-IV[£] | 46 | 11.30 | 41 | 11.50 |
| Durston et al. (2004) | Netherlands | Cross-sectional | Mixed | — | DSM-IV | 100 | 12.10 | DSM-IV | 100 | 11.60 | 100 | 10.70 |
| Fan et al. (2018) | Taiwan | Cross-sectional | — | Off | DSM-IV | 89 | 12.10 | DSM-IV | 70 | 13.00 | 78 | 11.80 |
| Faraone et al. (1993) – Boys | United States | Cross-sectional | — | — | DSM-III-R | 63 | 11.6 | DSM-III-R | 100 | 13.3 | 100 | 12.50 |
| Faraone et al. (1993) – Girls | United States | Cross-sectional | — | — | DSM-III-R | 63 | 11.6 | DSM-III-R | 0 | 13.0 | 0 | 13.00 |
| Gau and Huang (2014) | Taiwan | Cross-sectional | Mixed | Off | DSM-IV | 85 | 11.80 | DSM-IV | 37 | 12.30 | 73 | 12.60 |
| Gau and Shang (2010) | Taiwan | Cross-sectional | Mixed | Off | DSM-IV | 86 | 12.50 | DSM-IV | 41 | 12.50 | 73 | 12.60 |
| Goos et al. (2009) | United States | Cross-sectional | — | Off | DSM-IV | 79 | 9.10 | DSM-IV | 47 | 9.80 | 44 | 9.90 |
| Halperin and Gittelman (1982)[a] | United States | Cross-sectional cohort | — | — | DSM-II | 94 | 8.00 | — | 95 | 9.00 | — | — |
| Himpel et al. (2009) | Germany | Cross-sectional cohort | Combined | — | DSM-IV | 92 | 11.00 | DSM-IV[£] | 47 | 11.00 | 63 | 11.00 |
| Hwang-Gu and Gau (2015) | Taiwan | Cross-sectional | Mixed | Off | DSM-IV | 87 | 12.30 | DSM-IV | 31 | 12.70 | 81 | 12.70 |
| Jiang et al. (2020) | Netherlands | Cross-sectional cohort | — | — | DSM-IV | 61 | 14.61 | DSM-IV | 37 | 14.59 | 56 | 14.58 |
| Leon et al. (2024) | India | Cross-sectional cohort | — | Off | DSM-5 | 89 | 12.40 | DSM-5 | 69 | 11.80 | 77 | 12.60 |
| Luca et al. (2007)[a] | Canada | Cross-sectional | Mixed | — | DSM-IV | — | — | DSM-IV | — | — | — | — |
| Mulder et al. (2008) | Netherlands | Cross-sectional | Mixed | Off | DSM-IV | — | 14.90 | DSM-IV | — | 14.10 | 54 | 10.80 |
| Nikolas and Nigg (2015)[b] | United States | Cross-sectional | — | — | DSM-IV | — | — | DSM-IV | 32 | 11.30 | 54 | 10.80 |
| Oades and Christiansen (2008) | Germany | Cross-sectional cohort | Combined | Off | DSM-IV | 84 | 10.90 | DSM-IV[£] | 48 | 11.30 | 48 | 11.00 |
| Oades et al. (2010) | Germany | Cross-sectional | Combined | — | DSM-IV | 86 | 12.60 | — | 57 | 11.70 | 95 | 11.00 |
| Regalla et al. (2019) | Brazil | Cross-sectional | — | — | DSM-IV | 82 | 13.50 | DSM-IV | 44 | 14.80 | 51 | 14.70 |
| Reh et al. (2014) | Germany | Cross-sectional | Mixed | Off | DSM-IV | 78 | 9.20 | DSM-IV | 59 | 11.20 | 44 | 8.90 |
| Schachar et al. (2005) | Canada | Cross-sectional | Mixed | Off | DSM-IV | 67 | 8.50 | DSM-IV | 61 | 10.20 | 46 | 10.00 |
| Shang and Gau (2011) | Taiwan | Cross-sectional | — | Off | DSM-IV | 86 | 12.50 | DSM-IV | — | 12.20 | 73 | 12.60 |
| Shilon et al. (2012)[a] | Israel | Prospective case-control | Mixed | — | DSM-IV | 72 | 11.40 | DSM-IV | 72 | 12.00 | — | — |
| Slaats-Willemse et al. (2005)[c] | Netherlands | Cross-sectional | — | Off | DSM-IV | 92 | 12.29 | DSM-IV | 28 | 12.10 | 29 | 12.08 |
| Slaats-Willemse et al. (2003)[c] | Netherlands | Cross-sectional | — | Off | DSM-IV | 92 | 12.29 | DSM-IV | 28 | 12.10 | 29 | 12.08 |
| Slaats-Willemse et al. (2007)[c] | Netherlands | Cross-sectional | — | Off | DSM-IV | 92 | 12.29 | DSM-IV | 28 | 12.10 | 29 | 12.08 |
| van der Stigchel et al. (2007) | Netherlands | Prospective cohort | Combined | Off | DSM-IV | 100 | 10.90 | DSM-IV[£] | 100 | 10.10 | 100 | 8.90 |
| van Lieshout et al. (2019)[d] | Netherlands | Cross-sectional cohort | Combined | Off | DSM-IV | 82 | 11.44 | DSM-IV[£] | 41 | 11.21 | — | — |
| Wood et al. (2011) | Europe[†] | Cross-sectional cohort | Combined | Off | DSM-IV | 89 | 11.45 | DSM-IV[£] | 50 | 11.38 | 70 | 12.07 |

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; Med., medication phase; % Boys, percentage of boys included in the study; DSM, Diagnostic and Statistical Manual of Mental Disorders (DSM-II, second edition; DSM-III-R, third edition revised; DSM-IV, fourth edition).
Dashes (—) indicate information not available.

[§] Med. refers to the phase of the stimulant medication (on or off) during the cognitive assessment. Note that none of the studies clearly specified the inclusion of drug-naïve ADHD participants.

[†] The study includes participants from two or more European countries.

[£] The study is part of the IMAGE project. For more details see Brookes et al. (2006), Chen et al. (2008) and Müller et al. (2011).

[a] The following studies did not include non-sibling controls: Halperin and Gittelman (1982), Luca et al. (2007) and Shilon et al. (2012).

[b] Nikolas and Nigg (2015) did not provide cognitive test scores for participants with ADHD.

[c] Slaats-Willemse et al. (2005), Slaats-Willemse et al. (2003), and Slaats-Willemse et al. (2007) included the same study population but reported distinct cognitive test scores across the three studies.

[d] We did not extract data from controls because van Lieshout et al. (2019) included participants older than 18 years in the control group.

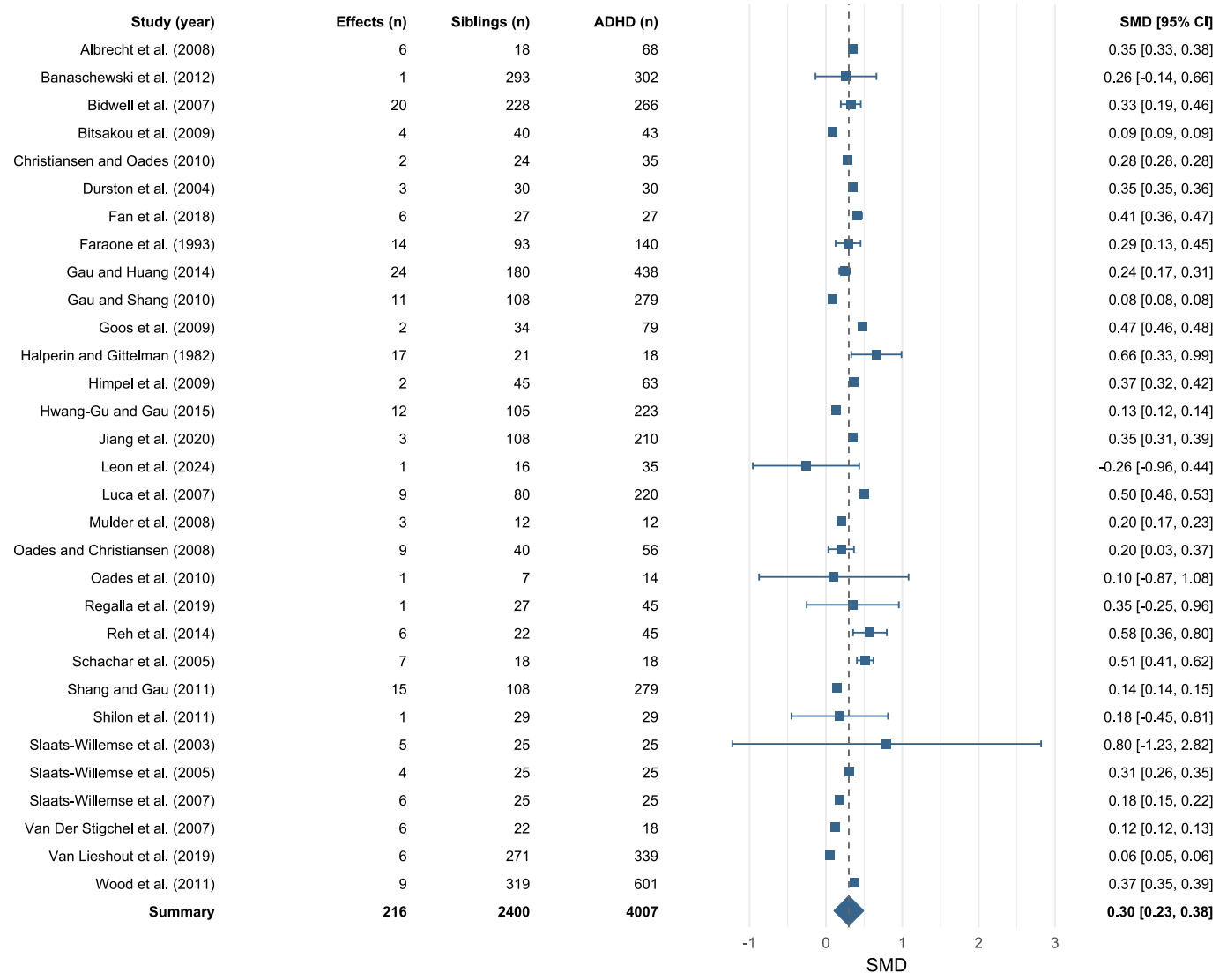



**Fig. 2.** This forest plot illustrates the estimated standardized mean difference (SMD) between siblings without versus with ADHD and its 95 % confidence interval (represented by horizontal lines) for each relevant study ($n = 31$). Positive SMD values indicate that siblings without ADHD outperformed their ADHD-affected siblings in cognitive outcomes. The dark blue diamond at the bottom of the plot summarizes the overall pooled SMD. For the 26 studies reporting two or more outcome measures, we applied cluster-robust variance estimation (CRVE) to derive a single summary effect size per study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

### 3.5. Meta-Analysis II: unaffected siblings vs. unrelated controls

The forest plot in Fig. 3 summarizes the relevant data on performance differences between non-ADHD siblings and unrelated controls, in addition to displaying the pooled SMD for each study included in the meta-analysis. In total, we computed 188 unique effect sizes from 27 studies, and the final meta-analysis sample involved 2094 siblings without ADHD and 2466 unrelated controls. The overall pooled SMD (−0.28, 95 % CI [−0.35, −0.21], $p < .001$) revealed that the cognitive performance of siblings without ADHD falls 0.28 standard deviations below the average of the control group. The amount of within-study heterogeneity (level 2: $\tau^2_2 = 0.0395$, $I^2_2 = 56.97$) was nearly four times greater than the estimated between-study heterogeneity (level 3: $\tau^2_3 = 0.0117$, $I^2_3 = 16.95$), meaning that over half of the total variation was attributable to within-study heterogeneity. In the sensitivity analysis, the removal of an outlying residual did not significantly reduce the within-study ($\tau^2_2 = 0.0327$, $I^2_2 = 49.79$) or between-study ($\tau^2_3 = 0.0150$, $I^2_3 = 22.80$) heterogeneity and yielded a similar SMD (−0.27, 95 % CI [−0.34, −0.20], $p < .001$). Consequently, the outlying residual was retained in the meta-analysis.

Regarding subgroup analyses summarizing data of unaffected siblings vs. unrelated controls by major cognitive domain categories (Table 3), the pooled SMDs demonstrate that siblings without ADHD exhibited significantly inferior performance relative to the comparison group in all four major cognitive domain categories under study: intellectual functioning (−0.34, 95 % CI [−0.50, −0.18], $p = .002$), attention and perceptual-temporal processing (−0.26, 95 % CI [−0.35, −0.16], $p < .001$), memory (−0.24, 95 % CI [−0.37, −0.10], $p = .021$), and executive functions (−0.28, 95 % CI [−0.38, −0.18], $p = .001$). Together, these effects suggest that siblings without ADHD tend to present with small-to-moderate difficulties in cognition compared to controls. The Wald-type test ($QM_{[3]} = 1.66$, $p = .647$) indicated that the difficulties did not appear to differ by major cognitive domain category, resulting in not rejecting the null hypothesis of equal between-cluster effect sizes. The computed within-cluster ($\tau^2_2 = 0.0387$, $I^2_2 = 54.01$) and between-cluster ($\tau^2_3 = 0.0149$; $I^2_3 = 20.82$) variance indicates that 54 % of the effect size variation was accounted by level-2 heterogeneity.

In the Supplementary Online Content, the univariate meta-

**Table 3**
Subgroup analyses by cognitive domain.

| Cognitive *subdomains* grouped by four cognitive *domains*[1] | Unaffected vs. ADHD-affected siblings[2] | | | | | Unaffected siblings vs. controls[3] | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Studies (*n*) | Effects (*n*) | SMD | 95 % CI | *p* Value | Studies (*n*) | Effects (*n*) | SMD | 95 % CI | *p* Value |
| Intellectual Functioning | | | | | | | | | | |
| Global Intelligence | 13 | 14 | 0.20 | [0.05, 0.35] | 0.016 | 10 | 11 | −0.40 | [−0.63, −0.18] | 0.003 |
| Scholastic Achievement | 4 | 9 | 0.56 | [0.10, 1.03] | 0.031 | 2 | 3 | −0.41 | [−2.14, 1.33] | 0.217 |
| Performance IQ | 7 | 17 | 0.27 | [0.09, 0.44] | 0.010 | 5 | 10 | −0.28 | [−0.49, −0.08] | 0.021 |
| Verbal IQ | 7 | 16 | 0.25 | [0.06, 0.44] | 0.017 | 5 | 9 | −0.33 | [−0.58, −0.09] | 0.021 |
| **Pooled SMD** | **15** | **56** | **0.29** | **[0.14, 0.44]** | **0.001** | **11** | **33** | **−0.34** | **[−0.50, −0.18]** | **0.002** |
| Attention and Perceptual-Temporal Processing | | | | | | | | | | |
| Attention and Psychomotor Speed | 7 | 40 | 0.30 | [0.18, 0.42] | 0.001 | 7 | 39 | −0.23 | [−0.37, −0.10] | 0.006 |
| Perceptual-Temporal Processing | 4 | 19 | 0.26 | [−0.01, 0.53] | 0.053 | 4 | 17 | −0.30 | [−0.47, −0.14] | 0.012 |
| **Pooled SMD** | **9** | **59** | **0.31** | **[0.19, 0.43]** | **< 0.001** | **9** | **56** | **−0.26** | **[−0.35, −0.16]** | **< 0.001** |
| Memory | | | | | | | | | | |
| Verbal Memory | 2 | 4 | 0.28 | [−0.23, 0.80] | 0.099 | 2 | 2 | −0.25 | [−0.77, 0.28] | 0.110 |
| Visuospatial Memory | 1 | 15 | 0.14 | [0.14, 0.15] | 0.003 | 1 | 15 | −0.22 | [−0.25, −0.19] | 0.006 |
| **Pooled SMD** | **3** | **19** | **0.21** | **[0.04, 0.38]** | **0.034** | **3** | **17** | **−0.24** | **[−0.37, −0.10]** | **0.021** |
| Executive Functions | | | | | | | | | | |
| Cognitive Flexibility | 4 | 21 | 0.26 | [0.02, 0.50] | 0.043 | 4 | 21 | −0.22 | [−0.36, −0.08] | 0.015 |
| Cognitive Inhibition | 12 | 43 | 0.41 | [0.24, 0.57] | < 0.001 | 12 | 44 | −0.27 | [−0.42, −0.11] | 0.004 |
| Planning | 1 | 3 | 0.26 | [0.00, 0.51] | 0.049 | 1 | 3 | −0.16 | [−0.34, 0.02] | 0.065 |
| Working Memory | 8 | 15 | 0.25 | [0.08, 0.43] | 0.013 | 7 | 14 | −0.37 | [−0.52, −0.23] | 0.001 |
| **Pooled SMD** | **19** | **82** | **0.32** | **[0.21, 0.43]** | **< 0.001** | **17** | **82** | **−0.28** | **[−0.38, −0.18]** | **< 0.001** |

Abbreviations: *n*, number; SMD, standardized mean difference; CI, confidence intervals.
[1] The outcome measures included under each cognitive domain are specified in Table 1.
[2] Combining the pooled SMDs: within-cluster heterogeneity: $\tau^2_2 = 0.0234$, $I^2_2 = 37.21$; between-cluster heterogeneity: $\tau^2_3 = 0.0235$, $I^2_3 = 37.46$.
[3] Combining the pooled SMDs: within-cluster heterogeneity: $\tau^2_2 = 0.0387$, $I^2_2 = 54.01$; between-cluster heterogeneity: $\tau^2_3 = 0.0149$; $I^2_3 = 20.82$.

regression analyses comparing unaffected siblings to controls (eTable 7) indicated that the sample mean age ($\beta = 0.052$, 95 % CI [−0.022, 0.126], $p = .141$), the percentage of boys ($\beta = 0.001$, 95 % CI [−0.008, 0.009], $p = .813$), and the year of publication ($\beta = 0.004$, 95 % CI [−0.012, 0.020], $p = .494$) showed no statistically significant associations with the effect size heterogeneity.

### 3.6. Risk of small-study effects

The funnel plots in Fig. 4 show a fairly symmetrical distribution of effect sizes, indicating no substantial small-study effects for both meta-analyses. In addition, the robustness to small-study effects was statistically confirmed for both unaffected vs. ADHD-affected siblings (Egger's regression test: z = −0.01, $p = .989$; 3PSM: $\chi^2_{[1]} = 0.09$, $p = .769$) and unaffected siblings vs. controls (Egger's regression test: z = −0.03, $p = .972$; 3PSM: $\chi^2_{[1]} = 0.08$, $p = .779$).

## 4. Discussion

To the best of our knowledge, the present quantitative synthesis is the first attempt at summarizing statistical data on intellectual and cognitive performance differences among children and adolescents without ADHD, their ADHD-affected siblings, and unrelated controls without ADHD. The current findings demonstrated that the neurocognitive profile of siblings without ADHD fell between that of their ADHD-affected siblings and the comparison control group. More precisely, siblings without ADHD consistently outperformed their ADHD-affected siblings but achieved significantly lower intellectual and cognitive scores compared to unrelated controls, with just under one-third of a standard deviation differentiating the groups overall. We did not find statistically significant differences among the pooled effect sizes estimated for the major cognitive domain categories (intellectual functioning, attention and perceptual-temporal processing, memory, and executive functions), which each showed significant performance differences between siblings without ADHD and both ADHD-affected siblings and controls.

Regarding the lack of significant differences between the major cognitive domain categories, it is important to keep in mind that multiple neurocognitive processes are involved in effective performance of complex neuropsychological tasks, and this is especially true for tests designed to assess intelligence and academic achievement (Peng & Kievit, 2020). An inspection of the (sub)tests included in the intellectual functioning category (Table 1) reveals that these measures involve a variety of cognitive processes that can potentially account for the heterogeneity of intellectual performance among unaffected siblings, as noted in the Introduction. For example, the versions of the Wechsler Intelligence Scale for Children (WISC) were designed to assess multiple neuropsychological outcomes, not just overall IQ. Of note, while the Vocabulary and Comprehension subtests from WISC depend heavily upon verbal reasoning and memory, the Block Design and Object Assembly subtests demand visuospatial processing and motor planning. Similarly, performance on scholastic achievement tests engage complex cognitive abilities, including semantic memory, executive functions, temporal information processing, and mathematical reasoning. In this quantitative review, inclusion of tests that tap multiple cognitive functions, combined with some rather broad major cognitive domain categories, may have led to intermixing of performance strengths and performance weaknesses, thereby preventing dissociation and thus identification of particular cognitive difficulties.

The intermediate level of intellectual and cognitive performance observed in siblings without ADHD, relative to their siblings with ADHD and unrelated controls, may relate to findings from neurofunctional and neurostructural imaging studies. Notably, both ADHD-unaffected and ADHD-diagnosed siblings tend to display atypical patterns of brain

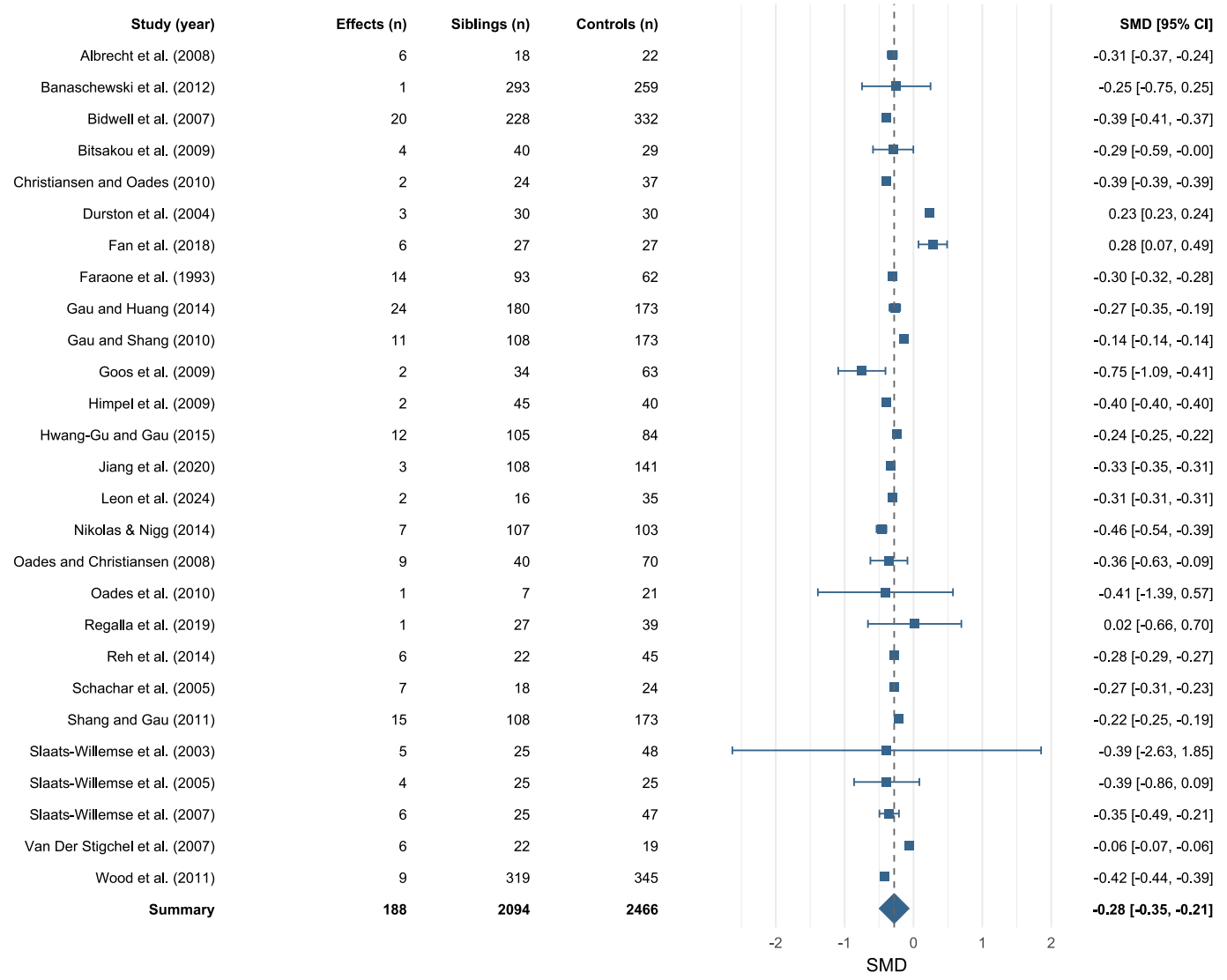



**Fig. 3.** This forest plot illustrates the estimated standardized mean difference (SMD) between siblings without ADHD and controls and its 95 % confidence interval (represented by horizontal lines) for each relevant study ($n = 27$). Negative SMD values indicate that siblings without ADHD showed inferior cognitive performance compared to controls. The dark blue diamond at the bottom of the plot summarizes overall pooled SMD. For the 24 studies reporting two or more outcome measures, we applied cluster-robust variance estimation (CRVE) to derive a single summary effect size per study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

activation compared to unrelated peers without ADHD (Bralten et al., 2016; Durston et al., 2004; Greven et al., 2015; Pironti et al., 2014; van Rooij et al., 2015). For instance, a prior neurofunctional analysis revealed that, relative to controls without ADHD, children without ADHD with a positive family history of ADHD showed significantly reduced frontal activation during working memory processing (*n*-back task) (Li, Motwani, et al., 2023). In addition, siblings of ADHD-affected adolescents and young adults have been reported to have gray matter volumes intermediate between those of their affected siblings and controls without ADHD, particularly in brain regions relevant to cognition (Bralten et al., 2016). Although conducted with adult samples, another study found that ADHD-affected individuals and their first-degree relatives without ADHD exhibit shared neurostructural distinctions (Pironti et al., 2014). Relatedly, siblings without ADHD showed lower region-specific and total cortical surface volumes relative to controls without ADHD (Hoogman et al., 2019). Taken together, it may be that the cognitive difficulties in siblings without ADHD across multiple neuropsychological domains as reported here are underpinned by neurofunctional and neurostructural imaging distinctions associated with familial ADHD. Importantly, in the literature on ADHD in children and adolescents, several of the brain areas showing distinctions have been implicated in the performance of neuropsychological measures designed to assess response inhibition (Barber et al., 2015; Booth et al., 2005; Huey-Ling et al., 2022) – an executive process for which difficulties are considered a core neuropsychological feature underlying ADHD (Barkley, 1997).

In their comprehensive and insightful review of the neuropsychological functioning of ADHD, Johnston and Chronis-Tuscano (2015) highlighted that the factors behind the reduced intellectual performance often observed in children with ADHD, compared to peers without ADHD, remain largely unexplained. On the other hand, twin studies suggest that poorer intellectual performance in ADHD, as indexed by lower IQ scores, seems largely driven by genetic influences (Kuntsi et al., 2004; Rommel et al., 2015). In relation to these previous considerations, findings from the current quantitative review add to the literature on intellectual functioning in ADHD by providing compelling evidence that siblings without ADHD are also more likely to achieve lower intellectual/academic scores compared to unrelated controls, albeit not to the

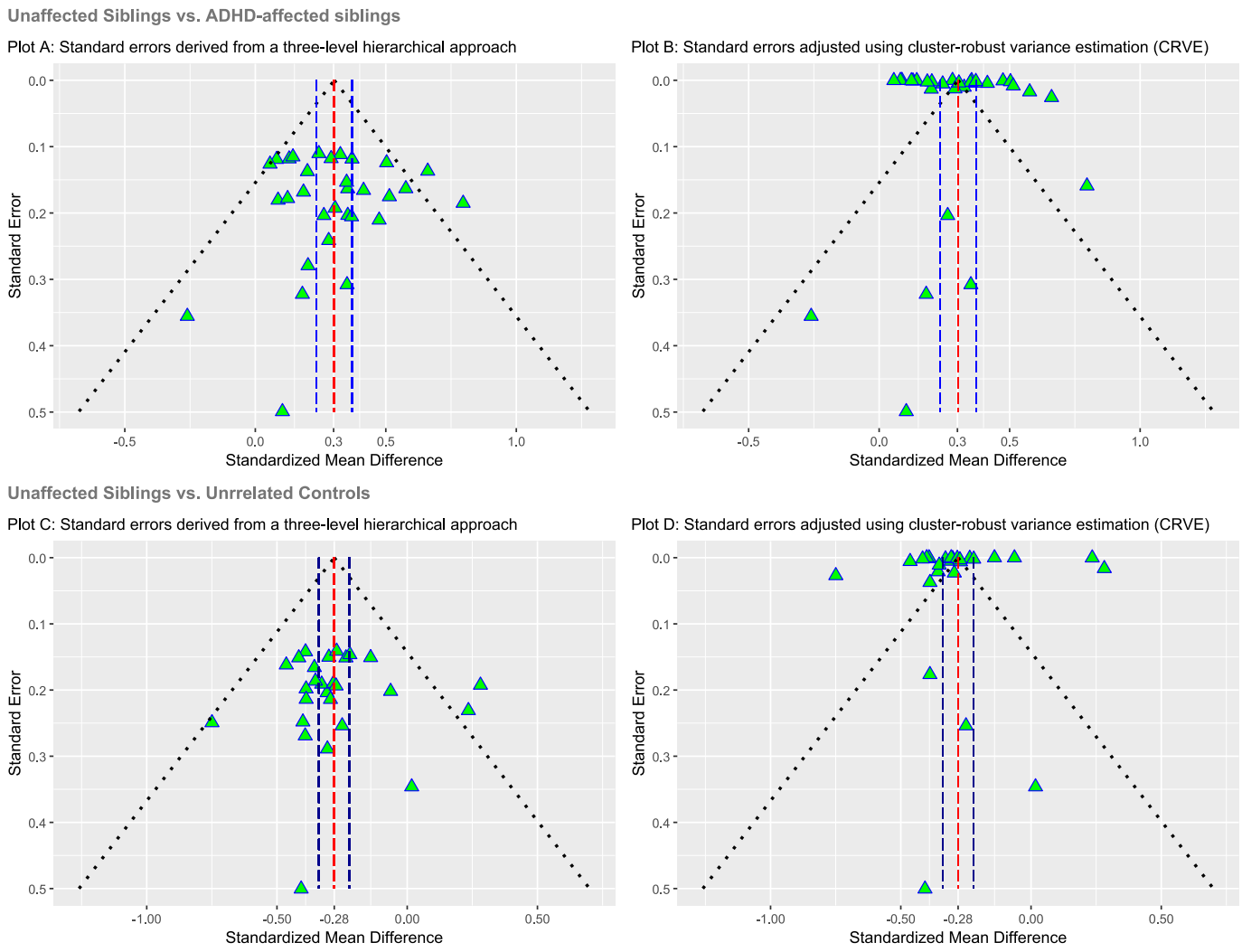



**Fig. 4.** Funnel plots for small-study effects. Green symbols represent the distribution of the pooled standardized mean difference (SMD) for each study considered in the meta-analysis on unaffected vs. ADHD-affected siblings ($n = 31$), as illustrated in Funnel Plots A and B, and in the meta-analysis on unaffected siblings vs. controls ($n = 27$), as depicted in Funnel Plots C and D. The dashed red line depicts the overall pooled SMD, and the dashed blue lines demarcate its 95 % confidence interval. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

extent of their ADHD-affected siblings.

Other factors that may be relevant to the findings discussed here pertain to the higher likelihood of families with ADHD sharing dysfunctional environments and experiencing more unstable households. For instance, an increasing number of studies have shown that children and adolescents with ADHD, and consequently their siblings, present with higher chances of having parents suffering from psychopathological disturbances, including depression (especially maternal depression), anxiety, conduct problems, antisocial behavior, personality disorders, and alcohol and drug-related abuse (Biederman et al., 1995; Johnston & Chronis-Tuscano, 2015; Kashdan et al., 2004; Margari et al., 2013; Wesseldijk et al., 2018). Compounding matters, coexisting with children and adolescents with ADHD is frequently a stress-generating experience (mainly for their siblings and parents), given that ADHD-related challenges (e.g., poor adherence to rules, difficulties with delayed gratification and routines, overactivity, erratic sleep patterns, and lack of motivation) often contribute to heightened within-family interpersonal conflict that leads to increased maternal distress, sibling rivalry and disrupted parent-child relationships, in addition to marital instability that in numerous circumstances culminates in separation and divorce (Angeline & Rathnasabapathy, 2023; Johnston & Chronis-Tuscano, 2015; Kendall et al., 2005).

Regarding the latter, a previous longitudinal analysis showed that parents of children with ADHD are 75 % more likely to dissolve their relationship within 10 years after the ADHD-affected child's birth (Kvist et al., 2013), while other studies have found that early-life stressful experiences such as neglect, household chaos, and trauma from parents' separation/divorce are associated with executive dysfunction and relevant difficulties in attentional processing and self-regulation (Andrews et al., 2021; Claussen et al., 2024; Lengua & Sandler, 1996). Moreover, past research on maternal parenting behavior and family functioning found that children and adolescents with ADHD and their siblings reported less perceived family support compared to unrelated peers without ADHD (Chang & Gau, 2017), which may be an important factor, given that family support has been shown to have a significant impact on academic performance (Li, Xue, et al., 2023) and also seems to be an important predictor of cognition and behavioral adjustment in childhood (Hughes et al., 2018). Adding further complexity, a less supportive home environment has been linked to higher rates of hyperactivity/impulsivity behaviors and oppositional problems in siblings without ADHD (Mulligan et al., 2013). Combined with genetic factors, these environmental factors may contribute to cognitive difficulties in siblings without ADHD.

### 4.1. Strengths

Notwithstanding that only studies published in English met the inclusion criteria, the comprehensive literature search we undertook considered studies published in English, Portuguese, and Spanish in order to minimize the risk of language-related bias, and thus this approach constitutes an important distinguishing feature of the present quantitative review. In addition, the current meta-analytic synthesis followed best-practice procedures to account for dependencies among effect sizes arising from studies reporting multiple outcomes. Specifically, we implemented three-level meta-analytic models with CRVE and carefully addressed potential sample overlap to ensure the inclusion of a unique effect size for each cognitive-intellectual test outcome. This thorough methodology is important towards ensuring the robustness of the results reported here.

### 4.2. Limitations

The present meta-analytical review comes with relevant drawbacks that must be pointed out. Regarding the search strategy, despite the multilingual approach strategy employed, we acknowledge that restricting the systematic search to studies published in English, Portuguese, and Spanish may still introduce selection bias. Relevant studies published in other languages may have been inadvertently excluded, particularly those from scientific journals not indexed in the online databases we used (PubMed/MEDLINE, Web of Science Core Collection, PsycINFO, and ProQuest). In relation to the study design, although observational studies have enabled relevant meta-analytical conclusions in medical and health sciences (Mueller et al., 2018; Muka et al., 2020; Paul & Leeflang, 2021), the nature of these studies precludes the inference of causal relationships.

Sample characteristics present additional limitations. First, only 10 studies clearly reported on the inclusion of ADHD siblings with the combined presentation, whereas the remaining studies either intermixed ADHD presentations or did not provide this information. Thus, we could not investigate whether between-group effect size differences depend on the condition presentation (combined, predominantly inattentive, or predominantly hyperactive/impulsive). Second, we were unable to conduct the planned investigation of potential effect size differences between children and adolescents due to only two studies (Bitsakou et al., 2009; Halperin & Gittelman, 1982) included in the quantitative syntheses providing sufficient data to distinguish participants aged 5 to 12 years (children) from those aged 13 to 18 years (adolescents). Although we did not find any statistically significant association through meta-regression analyses between the sample mean age of the sibling group without ADHD and the effect size variance, it may be that an age-group comparison would have had greater sensitivity to a developmental effect. Third, although the percentage of boys in meta-regression analyses did not show an association with the effect size variance, the well-known high prevalence of ADHD in boys (Arnett et al., 2015) indicates that future studies should further explore the potential influence of sex-based differences in cognition among siblings of ADHD-affected children and adolescents. Fourth, the included studies rarely reported detailed information on unaffected siblings beyond age, sex, and the statistics needed to compute effect sizes, and hence this limited reporting prevented a more nuanced psychosocial characterization of this group.

Additional methodological concerns involve heterogeneity in diagnostic and classification procedures. In particular, subthreshold ADHD traits could potentially have had a significant influence on the reported group differences. To shed light on this, ideally we would have included a subgroup analysis; however, the available information was insufficiently documented. For instance, only seven studies (Christiansen & Oades, 2010; Hwang-Gu & Gau, 2015; Jiang et al., 2020; Nikolas & Nigg, 2015; Slaats-Willemse et al., 2003; Slaats-Willemse et al., 2005; Slaats-Willemse et al., 2007) explicitly excluded or controlled for participants with subthreshold ADHD traits, and the operational criteria (e.g., number of symptoms/traits) were inconsistent across these studies. Furthermore, three of these seven investigations originated from the same research group (Slaats-Willemse et al., 2003; Slaats-Willemse et al., 2005; Slaats-Willemse et al., 2007), limiting generalizability.

Moreover, although most of the included studies employed standardized ADHD assessments (e.g., structured clinical interviews or validated rating scales), the specific diagnostic procedures varied considerably. Some studies applied comprehensive psychiatric evaluations, while others relied on brief screening tools or did not report procedures in detail. Variability in participant classification procedures may have contributed to heterogeneity in the meta-analytic findings and influenced the reported between-group differences. Finally, we feel that it is important to note that although we used the term controls, consistent with terminology used in most of the studies included in this quantitative review, terms such as neurotypical align better with recommendations pertaining to avoiding ableist language (French et al., 2025).

One final methodological consideration concerns the exclusion criteria applied across studies. We observed considerable heterogeneity in how these criteria were operationalized for probands with ADHD, unaffected siblings, and control groups, with reporting often too sparse or inconsistent to permit systematic coding. For example, some studies reported applying stringent exclusions for controls – such as prior history of neurodevelopmental (e.g., autism), neurological (e.g., head trauma), psychiatric (e.g., clinically significant mood or anxiety difficulties), and genetic conditions (e.g., fragile X syndrome, Down syndrome) – whereas others provided only a brief statement about excluding individuals with ADHD and with no behavioral problems at school. Such discrepancies could influence observed group differences. Future meta-analyses would benefit from more transparent and standardized reporting of exclusion criteria to enable their evaluation as a potential source of variability in effect sizes.

### 4.3. Implications

In line with studies demonstrating that siblings of children and adolescents with ADHD are at increased risk for psychopathological conditions and emotional difficulties (Orm & Fjermestad, 2021; Palacios-Cruz et al., 2014; Steinhausen et al., 2012), the neuropsychological outcomes presented in the current meta-analytic review highlight that siblings without ADHD warrant greater attention in future research to better elucidate their intellectual-cognitive profile. Within a family context, the needs of a sibling without ADHD may be overlooked by parents who may allocate more attention to the child with ADHD (King et al., 2016; Steinhausen et al., 2012), and hence identifying (neuro) psychological vulnerabilities in siblings without ADHD could inform and help guide not only parents but also the development of early intervention strategies. Early support with a family-focused approach could be offered to help mitigate unfavorable outcomes pertaining to the cognitive, academic, and social development of siblings without ADHD.

## 5. Conclusion

Findings from this quantitative review revealed that the intellectual and cognitive profile of children and adolescents not exhibiting clinically diagnosable behavioral characteristics of ADHD lies between that of their ADHD-diagnosed siblings and unrelated peers without ADHD. This intermediate neurocognitive profile was consistently observed across the four major cognitive domain categories we studied: intellectual functioning, attention and perceptual-temporal processing, memory, and executive functions. Ongoing prospective cohort studies on siblings of ADHD-affected children and adolescents are needed to provide insight into the persistence of these cognitive vulnerabilities into adulthood. In addition, the development of strategies and interventions aimed at enhancing the intellectual-cognitive abilities of siblings

without ADHD is needed in an effort to improve outcomes for these at-risk children and adolescents.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cpr.2025.102688.

## Data availability

Statistical coding and spreadsheets containing the collected data are available at the Open Science Framework (https://osf.io/sjwf3/)

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